

VisioSphere — Calculations & Tools

Live production configuration. Formulas as implemented, with the topic to search for each.

A1. CORE — feeds every detection

Calculation	Formula / rule	RRL search topic
Detection gating	DET_CONF 0.10 MIN_CONFIDENT_KPTS 5 / 17 POSE_CONF_THRESHOLD 0.45	Confidence thresholding in pose estimation; recall–precision trade under occlusion
Tracking	ByteTrack, two-stage association by detection score high 0.15 / low 0.05 / new 0.10 / buffer 90 / match 0.85	ByteTrack; low-score association; ID switches under rapid shape change
Keypoint smoothing	$x = 0.40 * x_{\text{raw}} + 0.60 * x_{\text{prev}}$ position frozen if conf < 0.15	Temporal smoothing of pose keypoints; EMA / low-pass; jitter in per-frame pose
Pixel-aspect correction	$k = (W_{\text{src}}/H_{\text{src}}) / (W_{\text{inf}}/H_{\text{inf}})$ $\tan(\text{true}) = k * \tan(\text{measured})$	Pixel aspect ratio; anisotropic scaling; viewpoint distortion in monocular measurement
Ghost rejection	$\text{torso_len} / \text{shoulder_width} \geq 0.60$	False-positive rejection in pose estimation; anthropometric proportion constraints

A2. FALL + LYING DOWN — shared geometry, one state machine

Calculation	Formula / rule	RRL search topic
Torso angle	$\theta = \text{degrees}(\text{atan2}(dx, dy)) $ shoulder-mid to hip-mid ; 0 = upright	Trunk inclination angle as fall indicator; body orientation from pose
Bbox aspect ratio	$\text{aspect} = h / w$ >1.5 upright , <0.9 fallen	Fall detection using bounding-box / shape aspect ratio
Height collapse	$\text{height_ratio} = \text{bbox_h} / \text{max_bbox_h}$ reference set on upright frames only	Silhouette height collapse; adaptive standing-height reference
Body extent ratio	$\text{extent} = \text{bbox_h} / \text{torso_len}$	Scale-invariant fall features; calibration-free anthropometric normalisation
Joint angles	$\arccos((v1.v2) / (v1 v2))$ cosine rule knee = (hip,knee,ankle) ; hip = (sh_mid,hip,knee)	Joint angles from 2D pose; seated vs supine posture classification
Standing-bend guard	$(\text{ankle_y} - \text{hip_y}) / \text{torso_len} \geq 1.2$ no ankle: $\text{torso_inv} \geq 0.30$ AND $\text{knee_drop} \geq 0.60$	Bending vs falling discrimination; ADL false positives
Furniture / floor zones	$\text{cv2.pointPolygonTest}(\text{polygon}, \text{support_point}) \geq 0$ support_point = hip midpoint	Context-aware fall detection; bed / chair region-of-interest priors
Path A (horizontal)	$\theta > 65$ AND $\text{aspect} < 0.85$ AND not bend	—
Path B (collapse)	$\text{height_ratio} < 0.50$ AND $\text{aspect} < 1.20$ AND $\text{extent} < 2.5$	—
Temporal confirmation	3 consecutive frames (STUMBLE 4, NORMAL 2)	Temporal filtering / multi-frame confirmation for false-alarm reduction
Size + edge gates	$\text{bbox_h} < 0.08 * 360 = 28.8$ px within 4 px of frame border	Minimum subject resolution for pose; truncation and occlusion limits

A3. FALL only — what separates it from lying down

Calculation	Formula / rule	RRL search topic
Peak angular velocity	$\text{rate} = (\theta_{\text{last}} - \theta_{\text{min}}) / (t_{\text{last}} - t_{\text{min}})$ 2.5 s window ; ≥ 45 deg/s = FALL	Trunk angular velocity; distinguishing falls from voluntary lie-down; speed of posture change
Time-bounded history	prune samples older than 30 s (not a frame count)	Frame-rate independence in video analysis

A4. LYING DOWN only

Calculation	Formula / rule	RRL search topic
Sustained confirmation	hold FALLEN 3.0 s before alerting	Dwell-time / duration threshold for lying confirmation; transient-pose rejection

A5. PROLONGED FALL

Calculation	Formula / rule	RRL search topic
Milestone escalation	30 / 120 / 300 / 900 / 1800 / 3600 s max 6 alerts ; measured from onset	“Long lie” — time on floor after a fall; alarm fatigue / alert burden

A6. INACTIVITY

Calculation	Formula / rule	RRL search topic
Centroid + EMA	$cx = x / \text{frame_w}$; $cx = 0.3 \cdot \text{new} + 0.7 \cdot \text{prev}$	Trajectory smoothing
Stationarity test	$\text{median}(v_i \text{ over } 2 \text{ s}) < 0.015$ $v_i = \text{hypot}(dcx, dcy) / dt$	Median filtering; robust statistics for noisy trajectories
Immobility threshold	1200 s ; milestones x1, 2, 4, 8 ; max 4	“Long lie” ; immobility detection in elder care; threshold duration
Posture qualifier	$\theta > 40 \text{ deg} \rightarrow \text{POSTURE variant}$ exempt if aspect > 1.5	Slumped / abnormal posture detection

A7. AGITATION

Calculation	Formula / rule	RRL search topic
Hands-on-head	$\text{wrist_y} < \text{nose_y}$ AND $ \text{wrist_x} - \text{nose_x} < 1.0 \cdot \text{shoulder_width}$	Hand-to-head gesture; agitation / BPSD markers in dementia
Wrist oscillation	reversal when $(y1-y0)(y2-y1) < 0$ AND $ y2-y1 > 0.15 \cdot \text{shoulder_width}$ ≥ 3 reversals in 4 s ; $\text{osc} = \min(1, n/5)$	Repetitive / stereotyped movement detection; zero-crossing or peak counting
Score + sustain	$0.70 + 0.60 \cdot \text{osc}$ (both hands) ; 0.15 (one) ; 0.25 (elbows) alert if > 0.25 held 15 s, 3 s grace, 300 s cooldown	Hysteresis in behavioural scoring; sustained-threshold alerting

A8. DELIVERY — applies to every alert

Calculation	Formula / rule	RRL search topic
IoU	$\text{inter} / (\text{area_a} + \text{area_b} - \text{inter})$ re-assoc $\geq 0.35 / 3 \text{ s}$; dedup $\geq 0.25 / 180 \text{ s}$	IoU data association in multi-object tracking
Circuit breaker	< 30 alerts per camera per rolling hour	Alarm fatigue; alert burden in clinical monitoring
Night gate	$\text{lum} = \text{mean}(\text{frame})$; engage < 18 , release > 22	Low-light degradation of detection; hysteresis (Schmitt) thresholding

B1. TOOLS — AI core, computer vision

Tool	Version	Role
YOLO11m-pose	20.9 M params	Person detection + 17 COCO keypoints per person.
OpenVINO	2026.3	Runs the model on Intel integrated graphics at FP16. Replaced NVIDIA CUDA.
Ultralytics	8.4	Model loading, frame preparation, tracking integration.
ByteTrack	bundled	Per-person track IDs. Tuned in bytetrack_fall.yaml.
PyTorch	CPU build	Ultralytics dependency. Not used for inference.
OpenCV	—	RTSP capture (TCP), resize to 640x360, polygon tests, overlays, clip writing.
NumPy	—	All geometry above: angles, ratios, velocities, median test.

B2. TOOLS — AI core, service layer

Tool	Version	Role
Flask + Flask-CORS	—	MJPEG feed, signed clip/upload endpoints, /status health check.
python-socketio	client	Sends alert events to the backend.
python-dotenv	—	Loads .env so every threshold above is tunable without code edits.
hmac + hashlib	stdlib	Short-lived signed tokens for stream and clip access.
FFmpeg	external	Transcodes clips to H.264 / yuv420p. Required — OpenCV cannot write H.264.
clip_server.py	systemd unit	Clip archive on a separate process from detection.

B3. TOOLS — backend and frontend

Tool	Version	Role
Node.js + Express	22.x / 5.2	REST API and application server.
Socket.IO	4.8	Receives alerts from the AI core; broadcasts to dashboards.
MongoDB + Mongoose	9.2	Incidents, residents, staff, assessments, audit logs.
JWT + bcryptjs	9.0 / 3.0	Authentication and password hashing.
Firebase Cloud Messaging	13.10	Push notifications to staff and guardian devices.
Resend	6.9	One-time passcodes for login and password reset.
node-cron / ExcelJS	4.2 / 4.4	Scheduled archiving; report and audit export.
Helmet / rate-limit	8.1 / 8.5	Security headers and request throttling.
React + Vite	19.2 / 7.3	Dashboard framework and build tool.
MUI + Tailwind	9.0 / 3.4	Interface components and layout styling.
Recharts / MUI X Charts	3.8 / 9.0	Incident statistics and trend charts.
Axios / socket.io-client	1.13 / 4.8	REST calls and live alert reception.
jsPDF + html2canvas	4.2 / 1.4	Client-side PDF report export.

B4. TOOLS — deployment and hardware

Tool	Version	Role
Dell OptiPlex 7010 Micro	i5-13500T	On-premise AI box. Intel UHD 770, 16 GB RAM.
Ubuntu	24.04	Operating system on the AI box.
systemd	—	Supervises detection and clip archive under /opt/visiosphere.
Cloudflare Tunnel	cloudflared	Exposes the AI box with no public IP.
Render	—	Hosts the Node backend.
Tapo TC65	RTSP / H.264	IP camera. Sub-stream feeds inference; RTSP forced to TCP.
Measured performance	12.18 fps	81.8 ms per frame. ~12 W idle, ~40 W processing. FP16.